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Hulling-milling values and nutritional evaluation of local varieties of rice (*Oryza sativa* L.) consumed by the tribal communities of Jharkhand

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Abstract : Rice (*Oryza sativa* L.) is one of the most important food crops grown all over the world. Jharkhand is inhabited by large numbers of tribal and other ethnic communities. Rice is their main food and they cultivate different types of rice varieties which possess many important characters, such as short maturation, biotic and abiotic resistance, good scent, low glycaemic index, antioxidant activity and grow easily without any input. In recent past local varieties are under negligence due to introduction of high yielding varieties. If this cultivation trend continues it is apprehended that valuable local varieties will be endangered and will be lost soon. However, one negative aspect of high yielding varieties is that they require much care, synthetic fertilizer and treatment with chemical pesticides and insecticides, which is not good for soil health and also for the environment. Therefore, there is an urgent need to focus and to take qualitative and quantitative improvement on the local varieties which are good source of valuable genes. Agronomical characters and nutritional values need investigation, before to start any breeding programme or *in vitro* studies, for improvement of local varieties. In the present investigation four local varieties of rice, namely Goda Dhan, Karhani, Pasara and Senegal Baba are identified for their nutrition and hulling milling values. The variety showing degree of milling percentage in order of merit are Senegal Baba (97%)followed by Karhani (95%), Pasara (90%) and Goda Dhan (86%). Moreover, the proximate nutritional values are variable for different varieties.

Key Words: *Oryza sativa*, local varieties, pesticides, insecticides, high yielding, agronomical

INTRODUCTION

Rice (*Oryza sativa* L.), belongs to family poaceae, is the most commonly grown species throughout the world. It is one of the most important staple crops which provide the source of carbohydrate for two - third of the world population. The grain is the most economically important part of the rice plant, and its endosperm is the final product which is consumed.

In the present investigation four local varieties of rice have been undertaken, namely Goda Dhan, Karhani, Pasara and Senegal Baba. All four varieties are different from one another in terms of agronomical traits; such as shape, size, scent, coloration, maturity, tolerance and yield capacity. Unfortunately, local varieties are under sheer negligence due to introduction of high yielding varieties. Therefore, there is great apprehension that if cultivation of high yielding varieties continues in view of the commercial value, local varieties despite carrying many good traits may soon be endangered and will be lost forever.

Therefore, there is an urgent need to protect, restore and conserve local varieties before it is too late. It is also desirable to bring qualitative and quantitative improvement in the local varieties so that these varieties may be the preferred choice for cultivation by the tribal communities of Jharkhand.

Plant breeding is one of the very old methods of improvement but it has certain limitations. On the other hand *in vitro* studies have great potential through haploid culture, protoplast culture, somaclonal variation and genetic transformation, which may be instrumental in improving

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the local varieties.

Before any improvement programme to be undertaken, screening of local varieties is one of the approaches for selection and identification of suitable variety. The first stage screening is the comparative study of agronomical characters; second stage is nutritional evaluation and third stage is callus induction and regeneration.

In the present investigation, therefore screening has been done in terms of hulling milling values and nutrition evaluation to identify good varieties for further studies.

MATERIALS AND METHODS

Materials

In the present investigation seeds of four local varieties of rice (*Oryza sativa* L.); Goda Dhan, Pasara, Senegal Baba and Karhani were collected from Local farmers and also from Birsa Agriculture University, Kanke, Ranchi (Fig. 1).

Methodology for hulling-milling

The grains were cleaned manually by removal of foreign matters, such as dirt, stones, immature rice etc. The manually cleaned rough rice was dehusked with rice dehuller and the brown rice obtained. The brown rice obtained from dehusker was milled for 30s in the friction mill. The resulting milled rice was separated manually or by grain grader into head rice and broken rice. The hulling-milling values are calculated as follows:

$$\text{Brown rice (\%)} = \frac{\text{Wt. of brown rice}}{\text{Wt. of rough rice}} \times 100$$

$$\text{Hull (\%)} = \frac{\text{Wt. of hull}}{\text{Wt. of rough rice}} \times 100$$

$$\text{Total milled rice (\%)} = - \frac{\text{Wt. of total milled rice}}{\text{Wt. of rough rice}} \times 100$$

$$\text{Degree of milling (\%)} = \frac{\text{Wt. of total milled rice}}{\text{Wt. of brown rice}} \times 100$$

Methodology for approximate estimation of nutritional ingredients

Nutritional evaluation of all the four varieties of rice were done for crude protein (CP), crude fiber (CF), total ash (TA), fat (Ether extract, EE) and moisture content according to methods of Association of Official Analytical Chemists.[1] The total percentage of carbohydrate content was determined by difference method.

Calcium content was estimated according to the modified method of Talapatra *et al.* (1940)[2], while phosphorus was determined by colorimetric method.[3]

RESULTS AND DISCUSSION

The result of the study showed varying approximation for rice varieties. The head rice recovery after hulling-milling values are shown in table 1. The highest proximate composition of carbohydrate and is followed by crude protein, moisture, crude fat, ash and fiber. In term of minerals only calcium and phosphorus had been estimated (Table **2).**

Hulling-Milling value

The chemical and nutritional quality of rice grain varies considerably and this may be attributed to genetic factors, environmental influences, fertilizer treatments, degree of milling and storage conditions. Major factors associated with the rice grain include cultivar and environmental variables, thickness of anatomical layers, size and shape of grain and the resistance of grains to breakage and abrasion. Major factors associated with milling process are processing methods, machines and milling conditions as they affect the degree of milling.[4]

No rice variety can commercially be successful unless it possesses high whole Kernel and total milled rice. Whole Kernel (head) yield is the quantity of intact whole kernel (including broken kernels, three quarters or more in length) of well milled obtainable from given quantities of rough rice (Paddy). Total milled rice yield include whole kernels (head) and other sizes of broken kernels obtainable from specified amounts of rough rice. The objective of rice milling is removal of hull, bran and germ with minimum breakage of endosperm.[5] In all varieties, unit mass of rice grain was found to decrease significantly for every level of processing by reason of husk bran and germ removal from paddy and brown rice respectively. In the present study recovery of head rice after hulling and milling process showed that var. Senegal Baba gave the best result (90%) followed by Karhani (88%), Pasra (86%) and Goda rice (85%).

Results also showed that milling results in reduction of the quantity of rice available for human nutrition by around 10 to 15 percent. There is similarity with the other

studies and results for different Indian milled rice varieties.[6,7,8]

According to Aroaullo *et al.* (1976)[9], the outermost tissues of the grain are commonly known as the husk and protect grain from undesirable conditions. The distinct space between the husk and caryopsis in dried grain filled with air and allows the grain to be dehusked without any or very little abrasion to pericarp. Furthermore, beneath the tegmen is a layer of tissue, several cells in thickness known as aleurone layer or bran. The shape of the cells of this layer is somewhat hexagonal to spherical. Thus, the hulling and milling operation cause the grain surface smoother that agrees with finding of Mohsenin (1978)[10], who claimed that the nature and the type of surface in contact play an important role in friction. The experimental results showed the effect of cultivars on the level of processing.

Several researchers[11,12,,13,14] deterermined some physical properties of rice grains for some varieties, levels of processing and moisture content.

Nutritional Analysis (Proximate composition of rice)

The nutritional value of rice cannot be generalized as it depends on numbers of parameter, such as genotype, irrigation and fertilizer applications. Rice is mostly considered a starchy staple food, which provides a large portion of dietary energy. Generally brown rice contains about 75 - 85 percent carbohydrates while processed rice shows even around 90 percent. Carbohydrate properties are therefore, an important factor determining the grain quality. Rice starch is usually digested quite rapidly, compared to other starch foods such as noodles, sweet potato or cassava.[15] The highest value of carbohydrate is shown bySenegal Baba **(78%)** followed by Karhani **(76%),** Pasara **(75%),** and lowest for Goda rice **(74%).**

Two local varieties, namely Senegal Baba and Pasra with broad grains are highly preferred by the poor people as they are slowly digested and give a long feeling of satiation after the ingestion. Moreover, due to slow digestion these varieties may be useful diet for type II diabetes patients. The carbohydrate content obtained in this study are similar to the values reported by other workers.[16,17,18]

The protein content of varieties undertaken for present investigation has been found ranging from 7- 12% which is high enough to meet the daily proteins requirement and generally higher than the high yielding varieties. Therefore, local varieties could be put under genetic maneuvering to improve the protein contain and it is a key factor influencing the eating quality of rice. This findings are similar to the earlier workers who analyzed the effect of cooking and soaking on physical characteristics, nutrient composition and sensory evaluation ***of*** indigenous and foreign rice varieties.[17,18,19]

In the present study fat content has been found varying at varietal level, ranging from 0.82-1.52%. The result of the present study are in line with the findings of Taira and Lee (1986)[20], who found range of crude fat from 0.50 to 2.23% in white and brown rice, respectively. However, this range is lower than the range obtained by Edeogu *et al.* (2007)[17]. This, difference may be attributed to the degree of milling. Milling of rice removes the outer layer of the grain where most of the fats are concentrated. It has been reported that those looking to reduce their fat and cholesterol intakes can turn to local varieties of rice because it contains only a trace of fat and no cholesterol.[21]

The crude fiber content ranged, in the undertaken varieties, from 0.008 to 0.05% which is much lower than the range (1.93 to 4.3%) obtained by Edeogu *et al.* (2007)[17]. It is similar to the mean value obtained by Sotelo *e t al.* (1990)[22]. Milling of rice generally decreases the fiber contents of rice.

It has been recommended by the experts to consume at least 25 gm of fiber every day to decrease the risk of chronic diseases. Fiber - rich foods help promote proper bowel function and reduce risk of developing intestinal disorders. One -half cup cooked white rice provides 0.3 gm of dietary fiber and one half cup cooked brown rice provides 1.8 gm of dietary fiber[23].

The moisture content of four undertaken varieties showed the range of 9-12%. In fact, high percentage moisture content affects the milling characteristics and the taste of cooked rice. Ebuehi and Oyewoie (2007)[17] reported that the moisture content of rice also affects its storage. The variation observed in moisture content among rice varieties may be attributed to differences in the genetic makeup and agronomic as well as climatic conditions.[22] The results of the present study are well in agreement

with the finding of different researchers.[23,,24,25] The high percentage moisture content may be attributed to low drying temperature and prolonged parboiling.[17,25]

The ash content of food sample gives an idea of the mineral elements present in the food sample. The ash content of the present local varieties ranged from 0.33 to 1.03%. Calcium content, which was maximum for Goda Dhan (0.152%), followed by Pasara (0.125%), Karhani (0.115), Senegal Baba (0.110), whereas phosphorus value is maximum in pasara (0.120) followed by Karhani (0.115), Pasara (0.110) and Goda Dhan (0.052).

Similar studies have been done by other workers[26,27,28] The differences might be as a result of fertilizer application, rate of parboiling and the amounts of soil nutrients. It is reported that as greater amount of rice bran are removed from grain during milling and polishing more vitamins and minerals are lost.

CONCLUSION

The local varieties of rice need immediate attention because they are under sheer negligence for long time due to introduction of high yielding varieties. Recently it has been realized that high yielding varieties are not good for soil health and environment because of the use of synthetic fertilizer and pesticides.

On this account and other important agronomical and nutrient features cultivation of local varieties need encouragement. Under rice improvement programme qualitative and quantitative improvement is an imperative. With this view four local varieties, namely Goda Dhan, Pasara, Karhani and Senegal Baba were studied with reference to hulling-milling and nutritional qualities out of which Senegal Baba and Karhani have been identified to be good enough for subjecting them to higher studies.

Table 1: Hulling -milling values of rice varieties

Varieties	Rough rice (gm)	Hull (%)	wn rice (gm)	Total milled rice (%)	Degree of milling (%)
Goda rice	100	27	70	60	86
Karhani	100	25	75	73	95
Pasra	100	29	71	65	90
Senegal Baba	100	28	72	70	97

Table 2: Comparative chart of nutritional analysis of rice varieties

Composition	Rice Varieties (%)			
	Gora	Karhani	Pasara	Senegal Baba
Moisture	10	10	9	12
C r u d e protein	10	09	12	0 7
Crude fat	1.52	1.25	1.38	0.82
Crude fiber	0.06	0.03	0.22	0.07
Ash	0 . 5 3	0.0.75	0.73	0.77
Carbohydrate	75	76	71	79
Calcium	0.152	0.094	0 . 1 2 5	0.123
Phosphorus	0.052	0.115	0.120	0.110

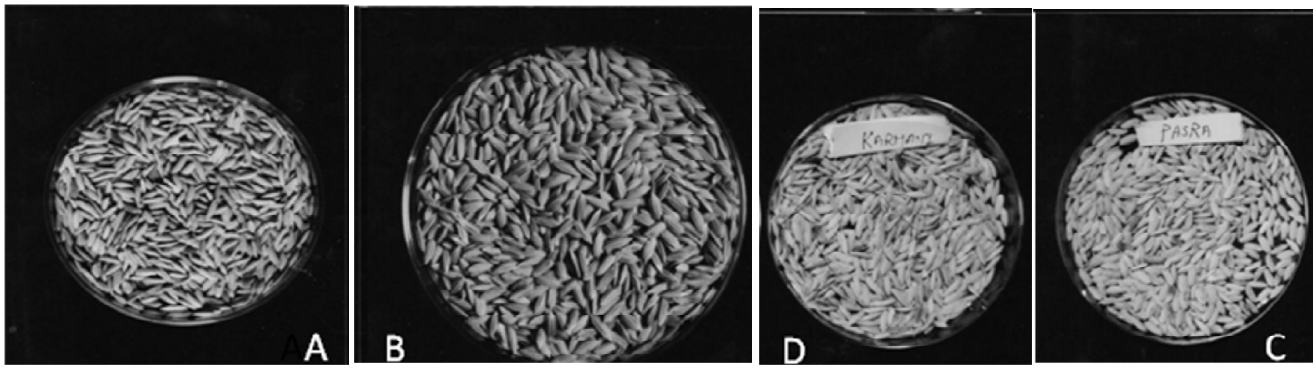


Fig. 1. Local varieties of Rice (*Oryza sativa* L.); A. Senegal Baba B. Goda Dhan C. Pasara, D. Karhani

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